NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4b Convection-scale AI and thunderstorm nowcasting

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 14:21 — 2026-11-09 06:19

Stage conclusion

Project report

Convection-scale AI and nowcasting (L / M / L2)

Goal

Determine whether convection-scale AI models at about 3 km (StormCast, WoFSCast, etc.) can directly produce thunderstorms and surface gusts. Also determine whether they can run batch forecasts/hindcasts for tens to hundreds of events. This is the only route in the thunderstorm direction that can scale up (the PI, 10-14). First-step deliverables: a verification table of whether public models run on a Colab A100; a verification table of 0–6 h forecasts for 5–10 downburst/severe thunderstorm events with station observations, against ASOS and HRRR; and, for the same events, AI/HRRR soundings vs observed soundings (the prerequisite gate for L2).

Current status

11-09 06:16: round 4 R4-S2 (StormCast V1 + method a, "any member" reading, 100 events + 100 near-threshold controls) finished. The frozen gate **passed** (hit difference −0.010 [−0.083, 0.060]; false-alarm difference +0.020 [−0.046, 0.085], with discriminating power; peak ratio 1.107 [1.034, 1.133]); the lead's independent recomputation matched digit for digit. Limits: hindcast configuration (ERA5 conditioning fields); the reading was chosen after seeing S2; Top-k capture is lower than HRRR; passing does not mean hits are non-inferior. Six-reviewer ruling (11-09 06:45): "pass" stands and is non-exploratory under §2.3 (another reading of preregistration row 5 is exploratory and disclosed alongside); absolute hits 19/100 vs. 20/100. Ruling recorded. Decision recorded.

11-09 06:13 EST: R4 segment B main sample finished 200/200 groups, each with 1 baseline + 10/10 members; VM1 and VM2 are both stopped and checked for leftovers; no units to rerun. Frozen r4_score_v4.py scoring: 100 events, 100 near-threshold controls, ΔH −0.010 [−0.083, 0.060], ΔF +0.020 [−0.046, 0.085], median peak ratio 1.107 [1.034, 1.133]; the four-branch overall verdict is "pass" (this configuration, sample and hindcast only; pending lead review). S2b was skipped under §6 because the frozen worker does not support 20-event segments; logged in DEVIATIONS. Draft generated by script. Cumulative use, estimated from S0 ≤3, VM1 43367 s, VM2 upper bound from 20:44:02 to 06:12:23 before shutdown, at 6.85 CU/h: ≤150.41 CU (an estimate, not a per-session bill). 11-08 02:40: the first round 4 S0 run was blocked by IEM rate limiting (HTTP 429, caused by our 8/10 concurrent threads). No controls were produced, no A100 was opened, and FROZEN was not appended. Decision: switch to serial rate-limited requests with on-disk caching and resume; stop on any failed request (do not record missing controls); then restart. Pairing rules unchanged (R4-D1). Extrapolations >3 CU move to the cluster. Assigned to the agent. Decision recorded.

10-28 lead re-review (6/6, consistent with the conclusion; this round was a duplicate submission that the lead had not noticed): failure upheld; negative result (limited to this configuration). Documentation completed: quantifying "false alarms have no discriminating power" is a post hoc addition (deviation D8). Report-only items: any-member hits 33/120 vs 33/120; maximum member peak ratio 1.096. Day-cluster intervals are not paired intervals. The 35 dBZ check covered only the event side. Finalized; decision recorded. 10-27 S2 conclusion finalized (lead checked: primary criteria match digit by digit). Report-only items with day-cluster intervals: HRRRCast heuristic BS difference −0.028 [−0.073,0.016], hit difference +0.108 [0.023,0.197], peak ratio 1.161. Only the false-alarm difference +0.100 [0.048,0.156] fails. 10-27 21:45 ruling after review by six independent reviewer models (6/6): the S2 conjunction **failed**, upheld. False alarms recorded as non-discriminating per §1.3. Stop category = **negative result (limited to the frozen configuration: StormCast V1 + method (a) gust diagnosis + P≥0.3/member median, 2 h)**. The remaining attribution was not separated: with the same AI fields under the HRRRCast heuristic definition, the BS/hit/peak point estimates pass the threshold, and false alarms are 16/120 vs 4/120 (report only, post hoc). S4 not run; S3 not opened. Testing the new definition requires a new preregistration and a new blind sample (about 180 CU); this was reported upward as a recommendation. Ruling issued; finalizing the draft assigned to the agent. 10-27 21:25 lead review: the false-alarm operator correction stands (deviation D7; consistent with preregistration §4.3; the diff changes only the operator and paths). The 2640 merged successful records are each traceable, with no duplicates or conflicts. All four point estimates were independently recomputed and match. The S2 frozen conjunction gate **failed** (BS, hits and peak ratio all fail, unrelated to the correction). Per preregistration §6, S4 is

not run and S3 does not open automatically. Candidate stop category: "negative result" for this configuration. Review by six independent reviewer models in progress; no external conclusion before the ruling. Decision recorded. 10-27 21:22 EDT: round 3 S2 groups 120–239 on VM2 all finished. The final checkpoint, logs and done marker were pulled back. Colab was stopped; no active sessions in the private or account session lists. After merging VM1, the frozen sample is 120 events + 120 controls, each with 10 successful members. CPU scoring: BS, hits and peak ratio fail; false alarms pass. So the S2 scale-up conjunction gate fails for this reported configuration. The full round used about 176.4 CU (allocated time × 6.77 CU/h estimate, not an independent bill), below the approved 185 CU. The frozen scoring script implemented the false-alarm difference with a wrong plus sign. The first-version result was kept, and a separately named corrected version uses a minus sign; this needs lead review. Conclusion draft and audit prepared. Not extrapolated to operational real-time forecasting or to general AI vs HRRR superiority. 10-27 13:00 EDT: round 3 S2 second VM; scientific criteria still not evaluated. 10-16 **On hold** (5/6 valid; ruling issued). What is on hold is scaling up the configuration "StormCast V1/CONUS + 10 m wind × 1.6677 gust proxy + post hoc ERA5 conditioning". Under the frozen thresholds, neither version met the scale-up standard (V1 20 events; CONUS 12 events, 53/54 groups). This is **not** "convection-scale AI is worse than HRRR". Hit differences are not significant (McNemar p 0.22–1.0). Criterion 1 passes fully under the ensemble-maximum definition. Counts of any member ≥25.7 m/s are no lower for AI than for HRRR GUST (V1 9/11/9 vs 8/8/6). V1 includes 7 already-seen cases, and the gust used only a constant coefficient. Why not continue: the current design can no longer answer "is it worth scaling up", and more samples would only spend money on a non-significant gap (CONUS about 700 CU per 100 events). Restart conditions (any one triggers a new project with newly preregistered criteria, not a direct resumption of scale-up): ① blind sample + thunderstorm wind verification + convective no-severe-wind controls + preregistered ensemble aggregation and probabilistic criteria + sample size estimate; ② a mixed-layer wind-profile gust diagnosis replacing the constant coefficient; ③ a public convection-scale model with native gust output or testable outside its training period; ④ a batch use that does not require station peaks (a change of goal, to be decided by the PI). Round 2 used about 212 CU in total. M is closed out; L2 not started.

Round 2 criteria (frozen 10-14; may not be relaxed afterward)

See the "Criteria" section. At each of 1–3 h, the median AI gust peak ratio must be ≥ 0.9 times that of HRRR GUST. For the ≥25.7 m/s discrimination (ensemble P≥0.3 counts as forecast), on event + no-event samples, the hit rate must be no lower than HRRR GUST and the false-alarm rate no higher than 1.5 times HRRR GUST. Under true GFS forecast conditioning, degradation relative to ERA5 must not exceed 20%. The gust coefficient is fixed at 1.6677 (2022 fit). Wide and narrow windows reported side by side, plus window-edge fractions and two HRRR baselines; persistence only in an appendix.

First-step criteria (written before the runs)

- Verification target: for each event, the ASOS station with the observed peak wind. For the model, take the maximum 10 m wind (and gust, if a gust variable exists) within ±25 km of that station and ±1 h of the

observed peak time. Baselines: the HRRR operational forecast at the same lead time (10 m wind and GUST) and persistence from the HRRR analysis at initialization.
- Metrics: peak ratio (model peak/observed peak), peak timing and position errors, and whether ≥25.7 m/s (50 kt) is forecast.
- "Worth continuing": at 1–6 h lead times, the AI peak ratio is no lower than 90% of HRRR's and beats persistence on at least 2/3 of events. In addition, inference for one event and one initialization takes ≤10 GPU minutes (the prerequisite for scaling up). If not met, state where it falls short; do not relax the criteria.
- Events must lie outside the training period of the weights used: StormCast V1 (trained 2018-07—2021-12, weights Apache-2.0) uses events after 2022; StormCast-CONUS (trained to 2025-09-30, weights OpenMDW-1.1) uses events after 2025-10.

How to scale up

The first-step script is written as a batch pipeline: "event list → automatic HRRR initial conditions → inference → automatic ASOS comparison". After passing the gate, extend to tens to hundreds of thunderstorm high-wind events with ASOS observations from SPC/Storm Events for hindcast statistics. The AIWP statistics from M give the predictable lead time of the environment, which decides whether L2 can do "forecasts" or only "hindcasts". The L2 CM1 ensemble runs on HPC; scale-up goes to the CM1 line or becomes a formal project.

Boundaries

- Hierarchy (root lead 10-14, maximum 5 levels): this node is a second-level node and will not open third-level sub-projects. When finer division is needed, use tasks within this node, or ask the parent node lead (thunderstorms and convection scale) to open a sibling second-level node.
- Only two hops remain downward: this node → agent → tools. Task briefs must not let tool agents (GPU, reviewer, literature agents, etc.) delegate further.
- The subtree list is not grouped by parent node; the hierarchy follows the project tree in the interface.
- Compute and data: Colab first, with no per-trial cap (aligned 10-14 with the parent project per the PI's words "use Colab more for now, no CU limit"; the old "≤20 CU per trial" is void). If a single trial is expected to exceed 100 CU, stop and report upward first. Always run colab stop after use; do not leave sessions idle. Only one executor per round. L round 2 stays as before: whole round ≤200 CU; report upward once at 120 CU cumulative; stop and report at 200 CU. HPC single job ≤50 SU, through resource requests. The L2 CM1 ensemble is run by the CM1 line; we only deliver profiles in input_sounding format.
- Where to compute and store (tightened by the PI 10-14): the workstation runs no computations and stores no computation data (10-15 01:59: drive C had 46 GB free). Solvers, inference, training, batch cases and large downloads all run and are stored on Colab or HPC clusters. The workstation holds only scripts, small results <200 MB per item, figures and reports. Pulled-back checkpoints stay small; large intermediates stay on Colab/the cluster. If >1 GB must be placed temporarily on the workstation, report

upward first. One cluster's group space is about 95% full, so new large data go preferably to the other cluster's work or scratch space; the first cluster defaults to a single node. Every task brief states where to compute and where to store results.

Decisions made

- 2026-10-14 literature review ruling: the two StormCast weight versions are treated separately. FuXi-Nowcast becomes a counterexample (its strong-gust threshold is worse than persistence). HRRRCast added as a candidate (public version 6 km, GUST is a post-processing diagnostic, pressure levels to 200 hPa, GFS boundaries); "only StormCast is usable" withdrawn. The Cook 10% wording narrowed; the Collins extrapolation withdrawn; novelty stated narrowly.
- 2026-10-14 project opened; first step entirely on Colab/workstation; ask the CM1 line before the CM1 part.

What we tried and results

Date	What was done	One-line result
10-16	Round 2 CONUS segment (B 46.2 + B2 small + B3 78.1 CU, estimated): 12 events from 2026 + 6 controls, 53/54 groups (one control 3 h group missing 1 member)	**Fails under the frozen criteria**. Wide-window AI/observed 1/2/3 h 0.67/0.74/0.68, HRRR GUST 0.79/0.82/0.81, HRRR10×coefficient 0.84/0.94/0.86; AI/HRRR GUST 0.85/0.90/0.84 → criterion 1 passes only at 2 h. Hits AI 1/1/0 vs HRRR GUST 4/2/3 (/12) → criterion 2 fails; false alarms 0 for both. The missing control group affects only false alarms and does not change the conclusion
10-15	Round 2 segment A: V1 + ERA5 conditioning, full sample of 20 events + 10 no-event controls × 1/2/3 h × 10 members (about 25.7 CU this session, rate estimate)	**Fails under the frozen criteria**. Wide-window peak ratio AI/HRRR GUST/HRRR10×coefficient: 1 h 0.68/0.73/0.90, 2 h 0.68/0.71/0.75, 3 h 0.64/0.75/0.85 → criterion 1 (AI ≥0.9×HRRR GUST) passes at 1 h (0.93) and 2 h (0.96), fails at 3 h (0.85). Criterion 2 hits AI 4/5/4 vs HRRR GUST 8/8/6 (/20), fails. Control false alarms 0/10 for both. Even the coefficient-scaled HRRR 10 m wind mostly beats AI. In the narrow window both reach only 0.4–0.5. Window-edge AI members 133–143/300

Date	What was done	One-line result
10-15	Round 2 gate segment: V1 runs the gust pipeline on the 7 first-step events (about 11.7 CU this session, rate × time estimate)	Pipeline and timing pass the gate: 173–312 s per group, member median 22 s (55 s estimated), about 0.3–0.6 CU/group. Scientific result (already-seen sample, not blind): wide-window AI/observed 1–3 h 0.51/0.59/0.62, HRRR GUST 0.75/0.73/0.77; AI hits 0/3, 0/3, 0/2 (HRRR GUST hit counts). Combined 10 events: AI/HRRR GUST median 1 h 0.84, 2 h 0.88, 3 h 1.01; ≥25.7 hits AI 3/2/2, HRRR GUST 6/6/4. In the narrow window both miss all 7 cases
10-14	Round 2 part ② V1/ERA5 (old rule ≤20 CU, counted in this round; account balance dropped 18.03 CU, an upper bound)	153-row checkpoint; 13 complete 10-member groups: 3 events (AMA, CDS, CID) × 1/2/3 h, 4 control groups (CDS 3 lead times, CID L1). 9/10 controls cross-checked with neighboring-station METAR <40 kt; CNK has no neighboring station. Interim summary (not a ruling): AI/observed median 1 h 0.79, 2 h 0.83, 3 h 0.77, HRRR GUST 0.91/0.92/0.79; ≥25.7 hits AI 3/3, 2/3, 2/3, HRRR GUST 3/3, 3/3, 2/3; control false alarms 0 for both. Compute: about 7 new groups used ≤18 CU in this part; at this rate the full ≥20+≥10 sample needs more than 200 CU, and per-group time must be measured and checked in the gate segment
10-14	Round 2 part ① HRRRCast GPU compatibility (about 1.1 CU)	Weights verified again; an isolated Python 3.10 + TensorFlow 2.15.1 environment detects the A100, but the kernel hangs when loading .keras, so event inference did not run. Decision: HRRRCast deferred (public version 6 km, training period covers 2023, environment not working); retry only after the V1 segment passes, then using events after 2024-04

Date	What was done	One-line result
10-14	Partial round 2 execution (pilot)	Two V1 cases (AMA 65 kt, CDS 62 kt) × 1/2/3 h × 10 members: with the gust proxy (AI 10 m wind × 1.6677, 2022 fit), wide-window AI/observed 0.77–1.04, comparable to HRRR GUST 0.72–0.92; P(≥25.7) AMA 1.0, CDS 0.3–0.7. CONUS runs on an A100 80GB (forward pass 97 s/2 steps, 20.6 GB); one OKC 2026 case at L1 AI/observed 0.66, P=0. CONUS pressure fields reach only p20hl (about 440 hPa). GFS-conditioned single-member smoke test passed. HRRRCast weights verified, but its training period 2021-03—2024-03 covers the 2023 events; preprocessing failed and no inference ran. The first session was reclaimed by the platform (pruned; reported to the platform). None of the three criteria were evaluated
10-14	HRRRCast availability check	Public weights exist (about 203 MB), but the public version is 6 km, pressure levels reach only 200 hPa, GUST is a post-processing diagnostic (factor 1.4), the weights have no separate license, and the training period was not checked; not run on Colab
10-14	First-step observation and baseline supplement: IEM ASOS 1-minute archive checked case by case for peaks; WoFSCast repository checked for weights	In 6/7 cases the 1-minute archive matches the reported peak in the same minute with the same value. OFK is missing in that minute (window maximum 53 kt; METAR PK WND records 56 kt), listed separately for recheck. Same-source comparison cannot rule out in-station QC removing peaks. ≥25.7 m/s hits: AI 0/28, HRRR GUST 8/28, HRRR 10 m 1/28. The WoFSCast repository has no weights and no release, and its configuration points to internal paths → unusable for now. The round 1 failure verdict is unchanged
10-14	M: AWS AIWP (Pangu, GraphCast Operational) on 40 ASOS ≥50 kt thunderstorm high-wind days in 2023–24, lead 12–120 h, against 13 ERA5 levels, baseline diurnal-cycle-aligned persistence, per-parameter criteria	DCAPE and 0–1 km shear pass at all five lead times; 0–6 km shear passes all five for Pangu and up to 72 h for GraphCast. MUCAPE/MLCAPE fail at all five lead times for both models (median error often below persistence, but they win only 14–25/40 events, less than 2/3). Verification time is the nearest 00/12 UTC, which can be up to 6 h from the wind peak; only the environment is verified, not gusts

Date	What was done	One-line result
10-14	First step: StormCast V1 (trained 2018-07—2021-12, central US 512×640) on 7 ASOS thunderstorm high-wind events from 2023–24 × 1/2/3/6 h initializations	Fails the preset threshold: AI/HRRR10 peak ratio median 1 h 0.85, 2 h 0.61 (3 h 1.11, 6 h 1.01); AI forecasts ≥25.7 m/s in 0/28; beats persistence in 6/7 events; ≤88 s per group, 100 events × 4 initializations about 6 GPU hours. Definition issue: AI and HRRR10 are both 10 m mean winds, reaching only 0.3–0.5 of ASOS gusts vs 0.75 for HRRR GUST; single member; conditioning fields from post hoc ERA5. Soundings: only 14 model levels up to about 440 hPa, so CAPE cannot be used for L2. WoFSCast has no public weights/initial conditions, FuXi-Nowcast has no weights, CorrDiff weights are restricted
10-14	Deep Research check	The two reports agree: no convection-scale AI model has been verified at stations for surface thunderstorm high winds (absent from searches); only StormCast is actually usable. Deep Research misattributed the HRRR high-wind bias to Fovell 2022; it is actually Collins 2024 WAF PDF p10 (gust −11.32 m/s, cold-season downslope winds). ASOS QC wrongly removes about 10% of convective peaks (PDF p1). No precedent for "AI soundings → CM1". Sent for review
10-14	Grok quick literature search + check of the StormCast model card and FuXi-Nowcast abstract	Current StormCast weights trained 2018-07–2025-09, full CONUS 3 km, OpenMDW-1.1 commercially usable, 10 m wind but no gust → verification events must be after 2025-10. FuXi-Nowcast has gusts but only for East China; no weights found. No precedent found for "AI soundings → CM1"

Parameter source table

Parameter	Value	Source or "assumption"	Where we looked
Verification neighborhood ±25 km/±1 h	25 km, 1 h	Assumption: about 8 grid points at 3 km, covering convective position errors; sensitivity: also report ±10 km/±0.5 h	Pending quick literature search
High-wind threshold	25.7 m/s (50 kt)	NWS severe wind standard	—

Open items and next steps

- 10-22 03:30 the VM1 disk filled up and S2 stopped at 120/240 (60 events + 60 paired controls; the 10 members of unit 120 must be rerun). About 91.3 CU used. Whole-round extrapolation is 180.1 CU at the effective unit price and 182.1 CU at the conservative unit price, above 180, so VM2 was not opened. Reported upward and requested approval up to 185 CU (corrected). Once approved, VM2 fixes the disk leak first and then resumes. Recorded.
- 10-21 19:32 lead re-estimate: about 37.2–37.5 CU used; units 20–47 done. Extrapolation 175.7–177.6 CU; with VM2 overhead 177.6–179.5 CU, still within 180. The agent's follow-up slots were taken by two expired follow-ups, so I will assign VM2 directly at my 10-22 04:26 follow-up.
- 10-21 19:30 S2 resumed: VM1 (opened 16:28) runs units 20–129 and reached about unit 46 at 19:23. The first 10 groups measured 6.77 CU/h in this session, i.e. 0.721 CU/group. Whole-round extrapolation 175.6–177.5 CU, excluding the second VM's overhead, leaving only about 2.5–4.4 CU of margin below 180. The keepalive's original keep-5 setting had automatically deleted 2 early non-final checkpoints; it is now set to keep 1000. The agent was assigned the wrap-up: re-estimate every 40 groups; after VM1 finishes, open VM2 for 130–239; then score, write the conclusion draft and report upward. I scheduled a backup follow-up at 10-22 04:26.
- 10-21 13:11 the first S2 batch of 20 groups finished (20/20 groups with 10/10 members; 387.7 s/group, 0.738 CU/group at 6.85 CU/h). Extrapolating, the full round needs about 178.9 CU, above 160, so the agent stopped and reported upward. The PI approved raising the cap to 180 CU. At 13:3x the agent was assigned to continue with groups 20–239. Per D6, two sequential VMs are used (about 23.7 h remaining, beyond the 24 h lifetime of one VM); both measured and conservative unit prices are reported.
- 10-21 the PI approved releasing S2 (Colab ≤160 CU); the platform registered the gate. At 09:5x the agent was assigned to run S2 (task brief); it is queued because the gateway delegation slots are full. After the first 20 groups, report the session name, CU per group and a whole-round re-estimate.
- 10-16 01:52 round 3 preregistration frozen:
- Finalized, with SHA256 recorded.
- Based on the review ruling and the decision record.
- The agent was assigned S0+S1 (Colab, ≤12 CU).
- 10-16 02:10 the agent returned the pre-screen (about 1.7 CU): tentatively 295 V1 and 127 CONUS candidates (grid-boundary checks and control pairing not yet done, so not a frozen sample); S1 not started. 02:20 follow-up assignment: after S0 wraps up, continue with S1; both together ≤12 CU.
- 10-16 02:28 returned: the two S0 jobs were estimated at about 3.4 CU at the A100 unit price, above the 3 CU segment estimate, so the agent stopped. Sample not frozen; S1 not started (grid checks confirmed all tentative candidates are inside the grid). I recorded this as deviation D1: segment CU figures are only budget estimates; S0 raised to ≤5, with S0+S1 still ≤12 in total. At 02:36 assigned S0 freezing and S1 in one go.
- 10-16 03:26 returned, stopped on an accounting issue: the CPU session was billed at the A100 rate of 6.85 CU/h and stopped at an estimated 9.8 CU. Eligible event pools were V1 301 and CONUS 132, with a provisional draw of 120/80; control pairing not yet done. I recorded this as deviation D2: CPU sessions are billed at the measured rate. At 03:36 assigned control pairing, sample freezing and S1.

- 10-16 04:56 verified: control pairing running in the background of a CPU session, 22/200, no missing controls so far; CPU measured about 0.08 CU/h, confirming D2 accounting. The agent scheduled a 07:45 follow-up to freeze the sample and run S1. I left a note requiring S1(a) to use all native levels of HRRR wrfnat.
- 10-16 10:49 S0 frozen: event and control lists, 120 V1 and 80 CONUS events, all 200 controls paired (eligible pools V1 301, CONUS 132; switching to 35 dBZ adds 0). 11:13 first S1 results: (a) 1.113 / r=0.915, (b) 1.00, (c) fail (AI Z1hl minus terrain is −26 to 39 m). The lead's recheck found the θv pressure unit wrong (Pa treated as hPa), so the first results were voided, recorded as D3. Heights now anchored to the lowest HRRR level, recorded as D4. At 11:32 the agent was assigned to rerun S1. Cumulative about 4.3 CU.
- 10-16 11:38 corrected S1 **passes fully**: (a) formula/operational fg10m median ratio 0.99999, r=0.982 (10 events, 4370 grid points × times); (b) using only StormCast's 14 levels, peak median ratio 1.000; (c) after anchoring, 860 columns in two events with NaN=0 and PBL top found in all. AI Z1hl drifts (−26 to 39 m), anchored per D4; columns with non-monotonic level heights handled per D5. Total about 6.4 CU. Reported upward; **S2 (V1 120+120, about 144 CU) awaits release by the parent project**.
- (Original plan) After S1 passes the gate, report upward. **S2 waits for release by the parent project and is not opened directly.**
- 10-16 the upper level ("AI meteorology") agreed with the direction and released S0+S1 (about 9 CU, conditional on the finalized version after review). The literature agent was assigned to build cards (Benjamin 2020 gust diagnosis original paper, radar convection thresholds). The finalization review by three independent reviewer models was sent, focusing on whether V1 80+80 has enough power and whether to raise to 100+ if there are more than 80 events. After both come back: rule, freeze the draft, release S0+S1.
- The literature agent returned (10-16). I spot-checked Benjamin 2020 PDF pages 14 and 16 against the original; accurate. Two changes at finalization: 1. The mixed-layer top criterion becomes θv(hl1, about 8 m) + 0.5 K (Benjamin 2020 PDF p. 14 / report p. 10), taking the maximum only within the PBL (PDF p. 16 / report p. 12). This removes the need for the q2m assumption. 2. The 40 dBZ radar threshold remains an "assumption": the literature supports only MRMS composite reflectivity >40 dBZ (Liu 2021, technical note p. 2). Roberts & Rutledge 2003 >35 dBZ (PDF p. 1 / journal p. 562) is used for sensitivity analysis. Cards built. After the S1 gate passes, report upward; **S2 is not opened directly** and is released by the parent project via the upper level.
- 10-16 round 3 preregistration draft written, **design only, no compute spent**. L remains on hold; release is decided by the upper level ("AI meteorology"). Key points of the draft:
- Blind sample: V1 uses events from 2022, 2024 and 2025-01–09; CONUS uses nationwide events after 2025-10. Controls are "convection without severe wind", paired 1:1 with events.
- Gusts switch to the HRRR mixed-layer diagnosis (Collins 2024 eq. (1), journal p. 708), requiring reruns that save levels hl1–hl20.
- Primary lead time 2 h; three primary criteria: Brier non-inferior, hits/false alarms non-inferior (margin 0.15), peak ratio ≥0.9.
- Sample size: V1 80 events + 80 controls; CONUS ≥50.
- Segments: S0 sample freeze + S1 pilot + S2 V1 blind test, about 105 CU; all five segments about 570 CU; S2 and each later segment must first be released by the upper level.

- Before finalization: the Benjamin 2020 paper and radar thresholds go to the literature agent for cards; the finalized version is sent for review under the frozen protocol.
- 10-15 12:50 the B session was stopped by its guard at 11:25 (46.19 CU, 230 rows, 20 complete groups). The new B2 session exited at startup with OptionalDependencyError and produced no new results (reported to the platform). At 12:58 the agent was assigned to open B3: install the optional dependencies as in the original B and resume from row 230 to complete all 54 groups. About 81 CU remained in the round (about 134 cumulative); stop and report at the limit.
- 10-15 parent project approval: run CONUS segment B to all 54 groups with a whole-round cap of 215 CU, replacing "stop at 200". Segment B resumes in place: no recomputation and no new session with the same name; above 215, stop and report upward. The agent was informed. On results: if it fails, send to six reviewers before putting it on hold; if it passes, report upward directly. Scale-up costs (CONUS about 700 CU, V1 about 140 CU) are recorded only and are not a basis for release.
- 10-15 10:42 recomputed from segment B's actual run time (upper-level request): the B session had run about 6.0 h, about 41 CU (at an estimated 6.85 CU/h), completing 17/54 groups. Wall time per group (1 event × 1 lead time × 10 members): 1 h about 14 min, 2 h about 20.5 min, 3 h about 27 min (more autoregressive steps run slower). The average is about 20.5 min ≈ 2.3 CU/group, about 1.6 times slower than the 13 min/group used at 05:20 (those 2 groups were 1 h only). Round 2 cumulative is about 83 + 41 ≈ 124 CU, past 120 (reported upward). The remaining 37 groups need about 87 CU, so the round is expected at about 210 CU and will hit the 200 cap. At 200 CU, stop and judge the roughly 49/54 completed groups under the frozen criteria, stating the sample size (events first; the gap falls on controls). The 05:20 statement "expected about 175, will not hit 200" is corrected per the measurements.
- Direct evidence for scaling up: CONUS on a single A100 takes about 20 minutes per group; 100 events × 3 lead times = 300 groups ≈ 100 GPU hours ≈ 700 CU. V1 takes 3–5 minutes per group; the same scale is about 20 GPU hours ≈ 140 CU.
- 10-15 05:20 round 2 accounting check (upper-level lead request):
- By segment (CU; "upper bound" = drop in shared account balance, including other concurrent sessions; "estimate" = session duration × about 6.85 CU/h): pilot ≤26.8 (upper bound); part ① HRRRCast about 1.1 (estimate); part ② V1 ≤18.0 (upper bound); gate segment about 11.7 (estimate); segment A about 25.7 (estimate); segment B to 05:10 about 3.3 (estimate). Total about 86.6.
- Definitions: the project page's old figure "upper bound about 58" is the cumulative before segment A (≤46 + 11.7). The "about 84/87" in reports = 58 + segment A 25.7 (+ segment B 3.3). Nothing was double-counted or missed; the two numbers refer to different times. The pilot and part ② are upper bounds, so the actual values are lower.
- Segment B estimate: about 13 minutes/group, about 1.65 CU/group (the first 2 groups include installation and weight loading, so actual per-group cost should be slightly lower); 54 groups about 89 CU. 3.3 counted by 05:10, about 86 remaining, plus about 2–3 CU for installation and loading in a resumed session. The round is expected at about 175 CU: it will pass 120 (report upward then) and will not hit 200. My 05:10 statement "likely to hit the 200 cap" was a miscalculation and is withdrawn.
- The 10:40 follow-up recomputes once more from segment B's actual per-group time and records it here.
- 10-15 05:10 CONUS segment (round 2 segment B) running: about 27 minutes for 2/54 groups; each CONUS group is far slower than V1. Upper-level lead approval: at 50 CU in one session, open a new session directly (resume from checkpoint); whole round ≤200 CU, report once past 120 CU. If not

finished at 200 CU, judge the completed groups under the frozen criteria and state the sample size. Report GPU minutes and CU per group, and estimate the cost of 100 events × 3 lead times.

- 10-15: files >100 KB under r2 (including the gate checkpoints) were moved to an archive on another cluster during the root lead's cleanup (with a list); the root lead will verify and pull them back once that cluster is restored. Resumption restores only from Colab; if the verdict depends only on these files, note "workstation checkpoints are on the archive cluster, not compared". This is the cause; the platform has been informed.
- Round 2 cumulative Colab (upper bound, including other sessions on the shared account): pilot ≤26.8 + part ① about 1.1 + part ② ≤18.0 ≈ ≤46 CU; report upward at 120 CU. Gate segment passed (10-15 00:26); cumulative upper bound about 46 + 11.7 ≈ 58 CU. Released segment A (V1 topped up to ≥20 events + ≥10 controls, ≤50 CU) and segment B (CONUS 2026 events + ≥6 controls, ≤50 CU). The GFS segment will be assigned only after some model passes criteria 1 and 2.
- Agent round 2 (task brief, queued after M): gust conversion, 10-member ensemble, ≥20 events + ≥10 no-event controls, true GFS forecast conditioning, trial run of StormCast-CONUS on events after 2025-10 with a check of its vertical levels; criteria written. 10-14 18:20 precheck: estimated about 163 CU, above the 100 CU gate, so it stopped (666 candidates). The lead had released 200 CU; on 10-14 the upper-level lead corrected this to ≤20 CU per segment (200 CU awaiting parent project approval). The 6 h lead was dropped and members batched for inference; order HRRRCast → V1 ERA5 → GFS conditioning → CONUS 2026 events.
- The upper-level lead added two interpretation points on 10-14 (passed to the agent; criteria unchanged): report the fraction of maxima at the window edge (≥0.9 × radius), with narrow and wide windows side by side. In round 1, the AI wide-window maximum fell at 20–25 km in about 20/28 groups, mostly background wind. The persistence baseline is too weak, so "6/7 better than persistence" is not evidence of skill; conclusions rest on the two baselines HRRR GUST and "HRRR 10 m × the same gust coefficient". Round 1 recheck: wide-window maxima at the window edge (≥22.5 km) AI 18/28, HRRR10 13/28, HRRR GUST 14/28; in the narrow window (10 km/±0.5 h), all three are 0/28 for ≥25.7 m/s.
- L2 soundings: StormCast V1 reaches only about 440 hPa and cannot feed CM1 directly; candidates switch to HRRR pressure-level products or the CONUS version (depending on the round 2 level check).
- 10-14 sounding supplement: the CM1 column-selection rule is written as a testable script (synthetic tests 2/2 pass). For the seven cases, RAOB vs ASOS peak time differences are 0.25–4.35 h (3 cases >3 h). The actual CM1 pre-convective columns have not been extracted: this needs the first ≥35 dBZ radar time per case, HRRR reflectivity and a land mask, and the raw profiles were not saved. The current sounding notes are reference only and cannot serve as the L2 handoff. Decision: L2 column extraction and input_sounding wait until round 2 judges "worth continuing" (using HRRR/HRRRCast native levels + NEXRAD/MRMS); not assigned now.
- M completed (see table above): global AI models forecast shear and DCAPE accurately 3–5 days ahead, while CAPE is unstable at 0.5–5 days. Implication for L2: hindcasts extract soundings from HRRR/ERA5 analyses; for forecasts CAPE is the weak point, relying only on short lead times of convection-scale models (round 2 HRRRCast has CAPE and 20 levels). No further M work.
- Grok quick search checked (see table above); Deep Research checked and the review ruled; HRRRCast added to round 2; the GitHub repository table is provided by the agent in the model list.
- 10-14 reply from the CM1 line: no duplication (they extract columns only from HRRR f00). The column-selection rule can be borrowed and was passed to the agent. Since 2022, only NIP 2022-06-03 and ALTU

2023-07-03 have measured peaks (both within StormCast's current training period, so not used as verification events). The small L2 ensemble (10–20 members) is run by the CM1 line; we deliver profiles in input_sounding format, scheduled after the cluster group space is restored and their sub-projects C/B start.

Literature basis

- PDF p3: CM1 downburst ensemble driven by 97 soundings (precedent for L2)
- Close precedents (no cards yet): CGsNet Xiao et al. 2023 GMD 10.5194/gmd-16-3611-2023; Lagerquist et al. 2017 WAF 10.1175/WAF-D-17-0038.1; FuXi-Nowcast arXiv 2512.08974v2 (station gust verification, 13.9/17.2 m/s worse than persistence); HRRRCast arXiv 2507.05658; Cook 2023 Meteorology 10.3390/meteorology2020017; Collins et al. 2024 WAF PDF p. 10
- Bolgiani 2020
- Parent project AI meteorology judgment table row 1b: StormCast arXiv 2408.10958, WoFSCast 10.1029/2024GL112383, FuXi-Nowcast arXiv 2512.08974, Das 2024 10.1038/s41612-024-00834-8 (no cards yet)

Notes from the PI

- 10-14: once 3 km scale is reached and favorable conditions are captured, CM1 downscaling can be considered. Precipitation is harder to forecast and is another hazard to consider later (P; not opened yet). The ultimate goal is forecasting and hindcasting at scale; for now, use Colab more, with no CU limit.

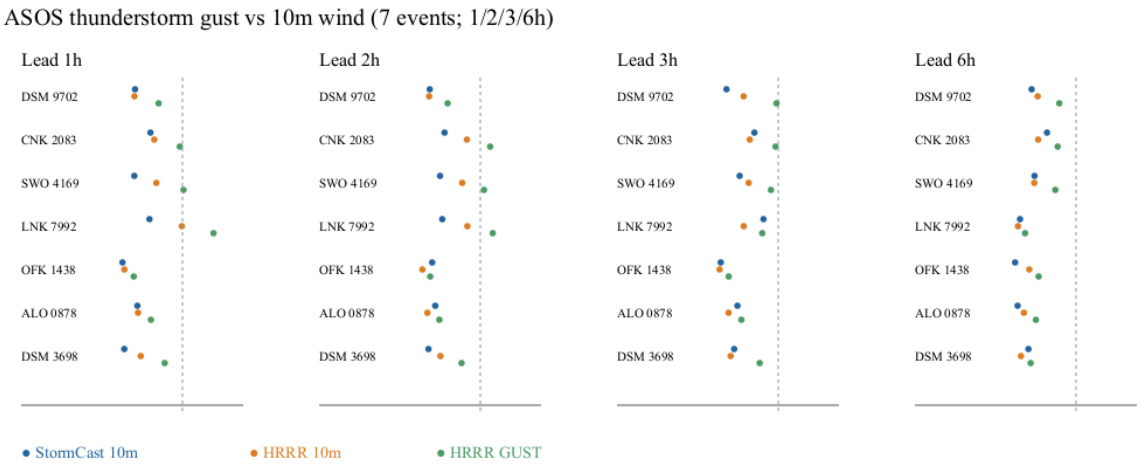
10-22 citation snowballing (the PI asked all sub-projects to redo snowballing)

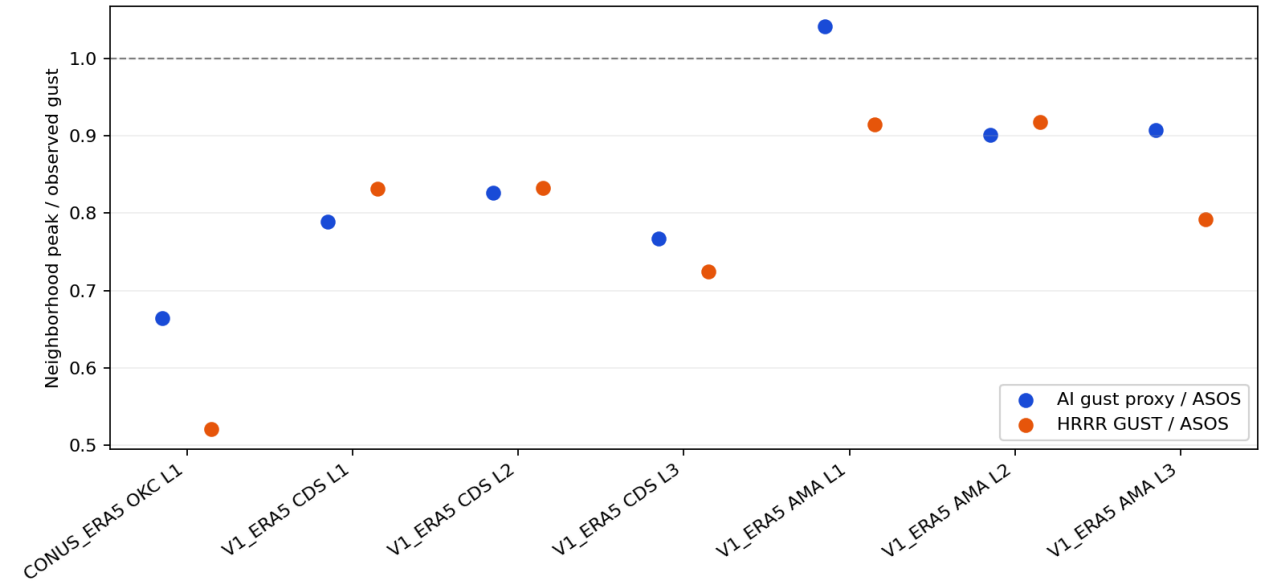
- Pool of 4625 papers (1733 after first screening; 884 found only by snowballing; includes Semantic Scholar). 13 papers selected for literature agent batch 7 (pending submission; automatic background retries).
- Notable at the title level: AIWP models directly producing severe convection forecasts (aies-d-25-0065.1); AI forecasts with Transformer post-processing for medium-range severe weather (aies-d-25-0045.1); a case comparison of AI models on storm impacts (arXiv:2511.01665). These may supply a literature contrast to "AI convective background fields only reach 12–24 h"; conclusions will not change before cards are built and original texts checked.

10-22 batch 7 verification

- **The conclusion must be narrowed**: "global AI models' convective background fields only reach 12–24 h" applies only to this line's three events and to local environmental quantities (CAPE, etc.). White 2026 (AIES, 10.1175/aies-d-25-0065.1) uses Pangu environment fields + random forests for 1–8 day severe convection (wind/hail/tornado) hazard probabilities. Tested on 2024, it has BSS>0 at all lead times and remains comparable to GEFS-MLP at day 7 (published PDF p.7; lead spot check matches). Hua 2026 is similar (3–8 days). Both forecast regional hazard occurrence probability, not local environmental quantities or station gusts. So they are not like-for-like counterexamples, but they show that "AI background fields + statistical post-processing" has skill at medium range.
- Other judgments of this line (the StormCast configuration did not meet the bar; v2 limited gap) are unchanged. ProbSevere v3, Jahan 2024 (WRF + ML gusts) and Hallgren 2024 (local hub-height wind trained on lidar) are adjacent precedents.

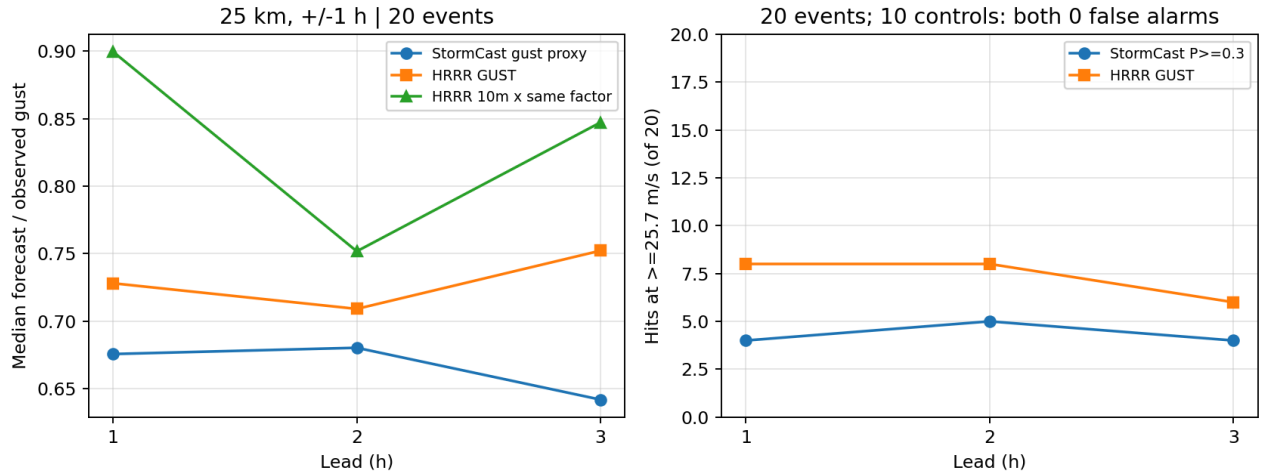
Key figures





Pilot only (2 V1 events + 1 CONUS event); NOAA Storm Events/IEM ASOS, AWS HRRR, ARCO ERA5, NVIDIA StormCast. Not r2 gate.

pilot peak ratio



Sources: NOAA HRRR / Storm Events, IEM ASOS; NVIDIA StormCast V1; ARCO-ERA5.
Gust proxy = 10 m speed x 1.6677 (2022 fit); retrospective ERA5, not operational.

v1 verdict